Estimating Infectious Disease Importation Risk during the 2026 FIFA World Cup

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Outline

The 2026 FIFA World Cup

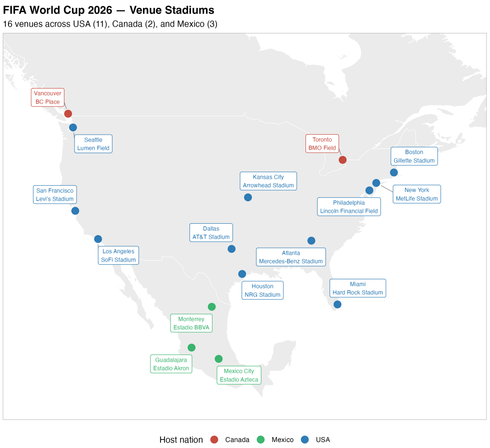
Key facts

- **48 nations** — largest World Cup in history
- **3 co-host countries**: USA, Mexico, Canada
- **16 venues** across 11 US cities, 2 Canadian, 3 Mexican
- Matches from **June 11 – July 19, 2026**
- Expected ~**85 million** international arrivals to the US in June 2026 alone (NTTO forecast)
- Tickets sold to fans from all 48 qualified nations

Why does this matter?

Mass gathering events accelerate the **importation** of infectious diseases into host cities. Early quantification helps public health authorities prioritise **surveillance** and **preparedness** by venue.

Venue cities — 16 stadiums across North America



Blue: 11 US venues **Red**: 2 Canadian venues **Green**: 3 Mexican venues. Suburb names mapped to metro areas (e.g. East Rutherford → New York).

Scientific question

What is the probability that at least one case of dengue, influenza, malaria, measles, or pertussis is **imported** *into each WC host city during June 2026 — and which source countries drive that risk?*

Five diseases studied

- **Dengue** — endemic in Latin America, SE Asia, Africa; strongly seasonal
- **Influenza** — Southern Hemisphere *winter peak* coincides with June tournament window
- **Malaria** — sub-Saharan Africa, SE Asia, Middle East
- **Measles** — resurgent globally; under-vaccinated pockets

Three nested models

1. **Baseline** — COR June 2024 $\times$ T-100 routing, no WC adjustment
2. **WC-adjusted** — 2026 country-level travel surge
3. **Schedule-driven** — fan routing by match schedule

The Poisson importation framework

We model disease importation as a **Poisson process**.

If $\Lambda_{v,d}$ is the *expected* number of imported cases of disease d arriving in city v, the probability of at least one importation is:

$$\boxed{P(X_{v,d} \geq 1) = 1 - e^{-\Lambda_{v,d}}} \qquad \text{(Poisson)}$$

where $\Lambda_{v,d} = \sum_c \lambda_{c,v,d}$ sums over all source countries c.

Each country-level contribution has the same structure across all three models:

$$\lambda_{c,v,d} = \underbrace{N_{c,v}}_{\text{arrivals}} \times \underbrace{I_{c,d}}_{\substack{\text{per-traveller}\\ \text{infection prob.}}} \times \underbrace{p_d}_{\substack{\text{travel while}\\ \text{infectious}}}$$

Disease-specific parameters

$$I_{c,d} = \rho_d \cdot \frac{C_{c,d}}{P_c}$$

- ρ_d — **under-reporting adjustment**: scales reported cases $C_{c,d}$ up to true incidence
- p_d — **probability of travelling while infectious**

Disease	ρ_d range	ρ_d (used)	p_d	Rationale for p_d
Dengue	0.06–0.26	**0.10**	0.50	Mild viremia; ∼40% viremic at arrival
Influenza	0.033–0.20	**0.10**	0.50	Moderate illness; many travel during early illness
Malaria	0.10–0.35	**0.20**	0.30	Severe illness; most febrile after return
Measles	0.40–0.80	**0.60**	0.05	Prostrating; only short pre-rash window
Pertussis	0.01–0.10	**0.10**	0.70	Catarrhal stage ≈ common cold; unrecognised

ρ_d ranges from: Bhatt et al. 2013 (dengue); WHO GISRS / Iuliano et al. 2018 (influenza); WHO WMR 2024 (malaria); Simons et al. 2012 (measles); McLaughlin et al. 2016 (pertussis). MC: 5,000 Uniform draws on each $(\rho_d,\ p_d)$ pair.

Data sources

Travel data

- **BTS T-100 International Segment**: nonstop flight segments at individual US airports (June 2023–2025) — country-level routing fractions for all **11 US venue cities**; used in all 3 models
- **COR I-94 data** (CBP): monthly arrivals by country of residence — June 2024 used as baseline (no WC growth for Model 1)
- **NTTO 2026 Forecast** (trade.gov): official 2026 projections for 12 top source markets

Disease data

- **Dengue**: WHO Global Dengue Surveillance (monthly, accessed Dec 2025)
- **Influenza**: WHO FluNet / GISRS global sentinel surveillance (ISO weeks 22–26, 2023–2025)
- **Malaria**: WHO Global Malaria Programme National Unit Data (2024 annual)
- **Measles**: WHO Immunization Data portal (annual, accessed Sep 2025)
- **Pertussis**: WHO Immunization Data portal (annual, accessed Dec 2025)

Monte Carlo uncertainty propagation

Both ρ_d and p_d span **one or more orders of magnitude** in the literature.

Approach: 5,000 independent draws from $\rho_d \sim \text{Uniform}(\rho_{\min}, \rho_{\max})$ and $p_d \sim \text{Uniform}(p_{\min}, p_{\max})$. Because $\Lambda \propto \rho_d \cdot p_d$, each draw rescales the central estimate by a scalar factor — *no recomputation needed*.

Why this matters

- Dengue/influenza 95% CI extends **upward**: risk may be underestimated
- Pertussis 95% CI extends **downward**: central estimate is a conservative upper bound ($\rho_c = \rho_{\max}$)
- Uncertainty is **integrated into every result**, not relegated to a sensitivity appendix

Three-model hierarchy

Model 1: Baseline
COR June 2024
T-100 routing
11 US venue cities

WC surge
(ϕ_c factors)

Model 2: WC-adjusted
NTTO + COR
2026 proj.
(ϕ_c growth factors)
11 US venue cities (T-100)

fan routing
by schedule

Model 3: Schedule
Same 2026 travel
WC fans $\rightarrow$ venue cities
11 US venue cities

No WC adjustment
country-level incidence
T-100 routing

Same routing &
incidence; only
travel volume changes

WC fans routed
by match schedule
to US venue cities

M1 $\rightarrow$ **M2**: isolates the WC travel surge (ϕ_c growth factors applied to COR 2024 base volumes).
M2 $\rightarrow$ **M3**: isolates schedule-based fan routing to US venue cities.

Model 1 — Baseline: equations

Why T-100 routing for the baseline? T-100 covers all **11 US venue cities** at country level, making it the only source that provides a consistent spatial grid across all three models.

Baseline arrivals (COR June 2024, no WC growth):

$$N_{c,v}^{\text{baseline}} = N_{c,\text{Jun 2024}}^{\text{COR}} \cdot f_{c,v}^{\text{T100}} \qquad \text{(baseline arrivals)}$$

$f_{c,v}^{\text{T100}}$: T-100 routing fraction (mean June 2023–2025), same in all three models, covering **all 11 US venue cities**.

Country-level incidence (shared across all 3 models):

$$I_{c,d} = \rho_d \cdot \frac{C_{c,d}}{P_c} \qquad \text{(incidence)}$$

Importation intensity:

$$\lambda_{c,v,d} = N_{c,v}^{\text{baseline}} \cdot I_{c,d} \cdot p_d \qquad \Lambda_{v,d} = \sum_c \lambda_{c,v,d} \qquad \text{(baseline } \lambda\text{)}$$

Model 2 — WC-adjusted: travel volume construction

Tiered approach to 2026 travel volume:

Tier 1 — NTTO top-12 markets (Australia, Brazil, Canada, China, France, Germany, India, Italy, Japan, Mexico, South Korea, UK)

Growth factor $\phi_c = V_{c,2026}/V_{c,2024}$ from official NTTO Forecast Report. Already includes WC effect — *do not* add WC visitors on top.

Remaining countries (COR I-94, June 2024)

- WC-qualified: $\phi_{\text{WC}} = 85{,}017/72{,}390 = \mathbf{1.174}$ (aggregate NTTO WC uplift)
- Non-qualified: $\phi_{\text{base}} = \mathbf{1.077}$ — data-driven from COR + T-100 June 2023–2025

$$N_{c,\,\text{Jun 2026}} = N^{\text{COR}}_{c,\,\text{Jun 2024}} \cdot \phi_c \qquad N^{2026}_{c,v} = N_{c,\,\text{Jun 2026}} \cdot f^{\text{T100}}_{c,v}$$

Model 2 — WC-adjusted: incidence and λ

Country-level incidence (more precise than regional):

$$I_{c,d} = \rho_d \cdot \frac{C_{c,d}}{P_c} \qquad \text{(country incidence)}$$

Importation intensity:

$$\lambda_{c,v,d} = N_{c,v}^{2026} \cdot I_{c,d} \cdot p_d \qquad \Lambda_{v,d} = \sum_c \lambda_{c,v,d} \qquad \text{(WC-adj } \lambda\text{)}$$

What changes from Model 1:

- Travel volume reflects the WC surge (ϕ_c growth factors applied to COR June 2024 base volumes)
- Routing fractions, city coverage, disease incidence, ρ_d and p_d are **unchanged** from Model 1

$\Rightarrow$ M1 $\rightarrow$ M2 isolates the WC travel surge alone

Model 3 — Schedule-driven: the core idea

Problem with Models 1 & 2
T-100 routing fractions reflect habitual tourist flows; WC fans travel to the specific cities where their team plays.

Reality: fans from Brazil go to *Houston* (where Brazil plays), not to Boston or Philadelphia.

Solution: separate the WC-specific travel increment from background tourism, and route fans using the match schedule.

Travel decomposition

$$N_c^{\mathrm{WC}} = N_{c,24}^{\mathrm{COR}} \cdot \max\big(0,\ \phi_c - \phi_{\mathrm{base}}\big)$$

$$N_c^{\mathrm{bg}} = N_{c,\mathrm{Jun\ 2026}} - N_c^{\mathrm{WC}}$$

For WC-qualified countries:
WC increment $\approx$ 8.3% of projected arrivals

$$\frac{1.174 - 1.077}{1.174} \approx 0.083$$

Model 3 — Schedule-driven: routing equations

Schedule-based WC fan routing fraction:

$$\omega_{c,v} = \frac{g_{c,v}}{G_c} \qquad \text{(schedule routing)}$$

$g_{c,v}$ = games played by country c's team at venue city v; G_c = total known group-stage games.

Arrival matrices:

$$N_{c,v}^{\mathrm{WC}} = N_c^{\mathrm{WC}} \cdot \omega_{c,v} \qquad \text{(WC fans} \rightarrow \text{venue cities)}$$

$$N_{c,v}^{\mathrm{bg}} = N_c^{\mathrm{bg}} \cdot f_{c,v}^{\mathrm{T100}} \qquad \text{(background} \rightarrow \text{all 11 US cities via T-100)}$$

Combined importation intensity for venue city v:

$$\Lambda_{v,d} = \underbrace{\sum_{c \in \mathcal{Q}} N_{c,v}^{\mathrm{WC}} \cdot I_{c,d} \cdot p_d}_{\text{WC fans}} + \underbrace{\sum_{c} N_{c,v}^{\mathrm{bg}} \cdot I_{c,d} \cdot p_d}_{\text{background}} \qquad \text{(schedule } \lambda\text{)}$$

Model 3 — Venue coverage

City	State
New York	New York
Dallas	Texas
Houston	Texas
Philadelphia	Pennsylvania
Boston	Massachusetts
Los Angeles	California
Atlanta	Georgia
Kansas City	Missouri*
Miami	Florida
San Francisco	California
Seattle	Washington

* Near-zero T-100 fractions; mainly WC-fan stream

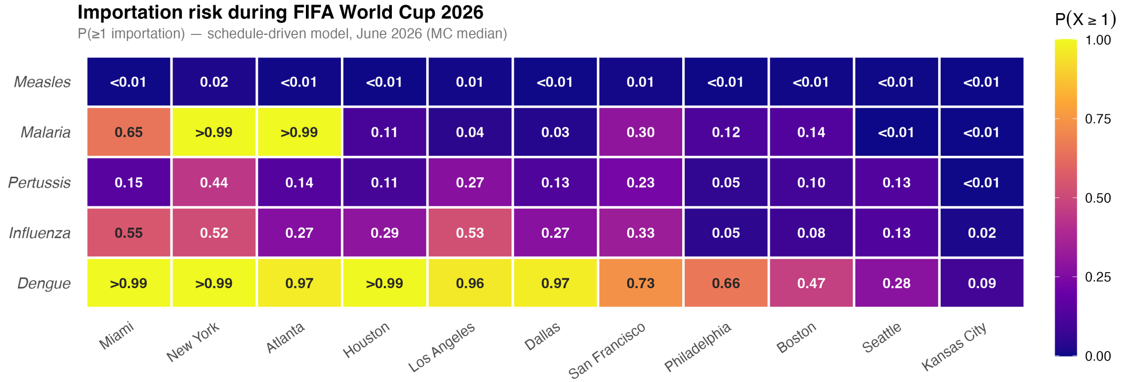
All 11 cities covered by T-100 routing fractions

Key insight

All 11 US venue cities have both WC-fan and background travel streams — T-100 provides routing fractions for all.

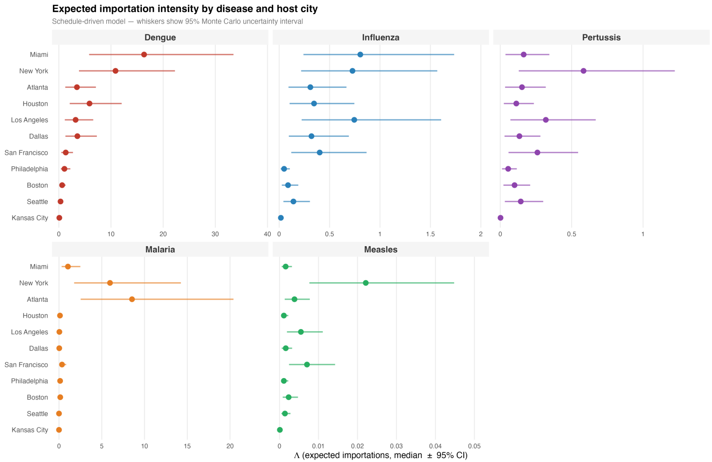
England + Scotland issue:
Both map to "United Kingdom" in COR data. Routing fractions sum to 1 — no double-counting.

Importation risk overview — all five diseases



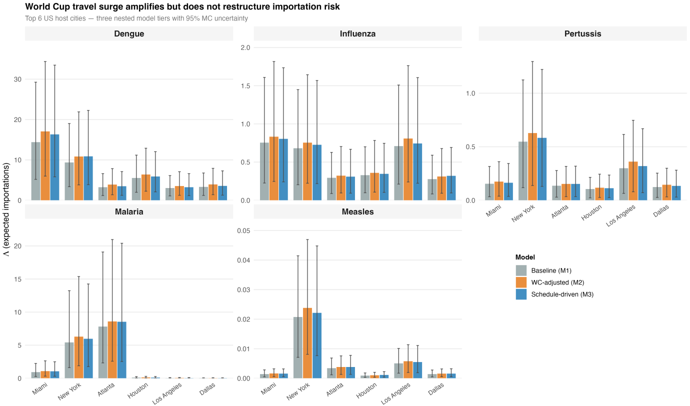
$P(\geq 1)$ under the schedule-driven model (MC median, 5,000 draws). Dengue is near-certain at every venue; measles risk is lowest. Cities ordered by decreasing aggregate importation intensity.

Importation intensity with 95% uncertainty intervals



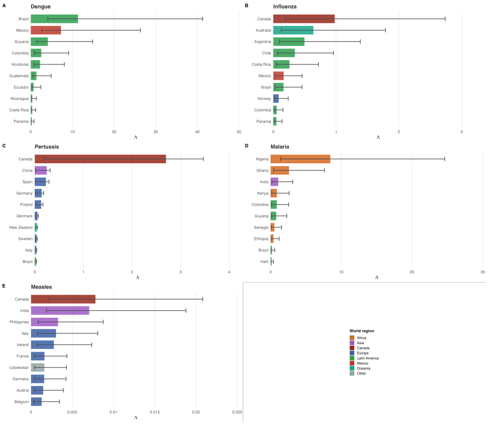
Point = MC median Λ; whiskers = 95% uncertainty interval (Uniform draws on ρ_d and p_d). Dengue is orders of magnitude above other diseases; pertussis CIs extend downward (central estimate is conservative).

WC travel surge amplifies but does not restructure risk



Grouped bars: Baseline (M1), WC-adjusted (M2), Schedule-driven (M3). Bars = MC median Λ; whiskers = 95% CI. M1→M2: near-uniform proportional surge across cities. M2→M3: city-specific divergence from schedule-based fan routing.

Top source countries by disease — with uncertainty



Top 10 source countries per disease (schedule-driven, all 11 US cities). Bars = MC median; whiskers = 95% CI. Colours = world region. Latin America: dengue & influenza; Africa: malaria; Europe: pertussis & measles.

Key findings

1. **Dengue importation is near-certain** ($P(\geq 1) > 0.99$ at every US venue city) — robust across the full MC parameter space.
2. **Influenza ranks second**, amplified by coincidence with the Southern Hemisphere winter peak during the June tournament window.
3. **The WC travel surge** (M1 $\rightarrow$ M2) amplifies risk *proportionally*; city rankings are stable.
4. **Schedule-driven routing** (M2 $\rightarrow$ M3) concentrates risk at specific US match venues; Kansas City gains meaningful risk almost entirely through the WC-fan stream.
5. **Background travel dominates**: the WC-fan increment is $\approx 8.3\%$ of projected arrivals for qualified countries.
6. **Country rankings are robust**: the top three contributors per disease are stable across the full MC parameter range.

What this model can (and cannot) offer

What it can offer

- A **transparent, reproducible baseline** for importation risk — built entirely from public data
- A natural **planning timeline**: M1 once cities are known; M2 when projections arrive; M3 once the schedule is confirmed
- Country $\times$ city risk breakdowns that can **inform** which cities and source streams to prioritise for surveillance
- A **quantified uncertainty range** (MC 95% CI) — shows where estimates are robust and where they are not

What it cannot replace

- Local context: vector presence, vaccination coverage, healthcare capacity
- Real-time surveillance — data lags mean this is a *planning* tool, not an operational alert system
- Onward transmission risk after importation
- Event-specific travel data (ticket sales, booking records)

The outputs are one transparent, reproducible input to preparedness planning — not a substitute for local epidemiological judgement.

Limitations

Parameter uncertainty

- **Monte Carlo uncertainty integrated**: 5,000 Uniform draws on (ρ_d, p_d) — 95% intervals reported for all main estimates
- Dengue/influenza CIs skew upward (risk may be understated); pertussis CIs skew downward (conservative upper bound)

Travel data coverage

- T-100 covers all 11 US venue cities across all models
- T-100 covers nonstop segments only; connecting passengers appear at entry city

Model scope

- Importation only — no secondary transmission
- Knockout-round games excluded (TBD opponents)
- Fan routing assumes proportional allocation by games; actual behaviour may differ

Data recency

- Disease data through late 2025; current outbreak situations may differ
- NTTO projections pre-date the event; actual arrival volumes will vary

Data that would most improve the model

Travel volume

- **Flight bookings** (OAG, Amadeus, IATA PaxIS) — direct country-to-city routing for June 2026
- **FIFA ticket sales** by country — most direct measure of WC fan travel; replaces NTTO decomposition
- **Previous WC entry records** (Brazil 2014, Russia 2018, Qatar 2022) — empirical validation of fan routing

Disease burden

- **Real-time surveillance** (ProMED, WHO DON, ECDC CDTR) — catch outbreaks after the 2025 data freeze
- **Age-stratified incidence** — travelers skew working-age; refines $I_{c,d}$
- **Vaccination coverage** (WHO/UNICEF) — adjust effective measles/pertussis incidence for traveler immunity

Local transmission

- **Ae. aegypti surveillance** (CDC ArboNET) — extend to local dengue outbreak probability
- **Hospital capacity** (AHA survey) — contextualise risk for health-system pre-positioning
- **Mobility data** (Meta, Google) — validate schedule-based fan routing empirically

Next steps

1. **Incorporate knockout rounds** — as the WC bracket is resolved, update the schedule-driven model with actual match assignments beyond the group stage
2. **Real-time surveillance linkage** — compare model predictions against actual case notifications from CDC and partner health agencies during the tournament
3. **Expand disease scope** — chikungunya (active 2025–2026 outbreak in Brazil & Argentina) and mumps are the highest-priority additions; both have accessible public data
4. **Domestic routing extension** — use DOT enplanement data to capture importation pressure at cities receiving onward domestic connections from T-100 gateway cities
5. **Extended routing data** — booking-level data (OAG, Amadeus) to refine city-level routing fractions beyond T-100 segment data

Thank you

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Code and data:

https://github.com/jdiestra1977/FIFA_worldCup_2026_risk